

When Does Lightweight Post-processing Help?

A Seven-system JJAS Rainfall Benchmark over India

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Abstract

Weekly rainfall guidance at subseasonal leads could support climate-adaptation planning, but comparisons across forecasting systems are easily confounded by initialization calendars, valid dates, grids, and climatologies. We align seven physics, data-driven, and hybrid systems on one 1.5-degree India grid and evaluate JJAS initializations at lead Weeks 1–6 against India Meteorological Department (IMD) rainfall. Initial fitting uses 2020–2023; 2024 selects tree counts; final models refit through 2024; and no 2025 data enter fitting or selection. A compact XGBoost adaptation of PiggyCast raises mean case-level spatial anomaly correlation from 0.307 for the equal-weight mean to 0.347 (paired difference 0.040, 95% moving-block interval [0.011, 0.070]) and reduces RMSE from 5.174 to 4.989 mm day⁻¹. The result is deliberately bounded: the interval against a per-lead 2024-RMSE-selected individual baseline overlaps zero; ACC gains occur at Weeks 1–4 but not Weeks 5–6; and a positive national mean hides a clear loss in east/northeast India. Forecast-conditioned mixing therefore improves a simple ensemble baseline, but is not uniformly superior. We release case-level scores, coverage, trained post-processors, uncertainty intervals, and provenance hashes so every reported numerical claim is auditable.

Keywords: subseasonal forecasting; Indian monsoon; precipitation; multi-model ensemble; post-processing

1 Introduction

The two-to-six-week range sits between weather prediction and seasonal outlooks. Skillful weekly rainfall guidance could inform agricultural advisories, reservoir preparedness, and flood-readiness decisions, but those uses require evidence at the relevant lead, season, and region. The WWRP/WCRP S2S database enabled multi-centre comparison [Vitart et al., 2017]; recent AI systems such as FuXi-S2S [Chen et al., 2024], NeuralGCM [Kochkov et al., 2024], and DLESyM [Cresswell-Clay et al., 2024] add architectural diversity. It remains unclear whether a small post-processor can exploit these heterogeneous forecasts for Indian monsoon rainfall without retraining the base systems.

Regional benchmark claims are fragile when forecasts do not share the same initializations, valid-day endpoints, spatial support, or anomaly reference. Test-year reuse can further turn an apparent gain into model selection. We therefore ask: *When does lightweight mixing improve common-grid JJAS rainfall guidance over India?*

We contribute: (1) a valid-date-aligned benchmark

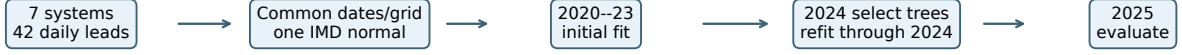
of seven systems for Weeks 1–6 nationally and over four IMD monsoon regions; (2) forecast-only, full, and location/calendar-only adaptations of PiggyCast [Kamani et al., 2025], alongside equal-weight, validation-weighted, and per-lead selected baselines; and (3) a 2025 evaluation with paired moving-block intervals and explicit claim gates. The useful result is not a claim of universal superiority: mixing improves equal weighting on average, while the lead-wise and regional failures identify where such a method should not be trusted without guardrails.

2 Benchmark and protocol

Forecast cohort and valid dates. We evaluate weekly ensemble-mean precipitation from CMA, DLESyM v0, ECMWF, FuXi-S2S, NCEP, NeuralGCM, and UKMO. Each retained initialization has 42 complete daily rates. Week 1 averages accumulation intervals ending at initialization+1 through +7 days; Week 6 averages +36 through +42. We use the common initialization intersection: 140 cases from 2020–2023 for fitting, 30 cases in 2024 for every choice, and 35 cases in 2025 for evaluation. The exact dates, ensemble sizes, and missing 2024 dates are reported in the appendix.

Truth, grid, and regions. IMD gauge-based daily rainfall [Pai et al., 2014] is conservatively aggregated to a 27 × 27, 1.5-degree grid. Cell weights combine spherical area, fractional national or regional overlap, and the observed IMD fraction. We use India and the northwest, central, south-peninsula, and east/northeast IMD monsoon regions. A fixed 31-day-window IMD 1991–2019 normal is averaged over each forecast’s exact seven valid dates and subtracted from both forecast and observation. Thus no model-specific climatology enters ACC. Finite negative precipitation increments from accumulated-field differencing are retained rather than selectively clipped.

Pre-specified split. Figure 1 summarizes the workflow. Model and feature sets, hyperparameters, anomaly construction, metrics, and uncertainty procedure were recorded before scoring 2025 within this workflow. The 2025 values are not used for normalization, calibration, fitting, early stopping, threshold choice, feature choice, or retry selection. This is an auditable workflow property, not a claim that the underlying source archives have never been viewed for other projects.



JJAS initializations; Weeks 1-6 use +1...+42 period endpoints; 35 common 2025 cases; no 2025 fitting or selection

Figure 1: Evaluation design. Season membership is set by initialization month, so later lead windows may verify after September. Within the released workflow, all fitting and selection precede the 2025 evaluation.

3 Methods

Baselines. The equal-weight mean averages the seven systems. A validation-weighted mean uses non-negative inverse-2024-RMSE weights separately by lead. A stronger pre-specified comparator selects the lowest-2024-RMSE individual system at each lead: FuXi-S2S at Week 1 and ECMWF at Weeks 2-6. We call this the *per-lead 2024-RMSE-selected baseline*; it is not one fixed model and was selected by RMSE even though ACC is our primary evaluation metric.

Forecast-conditioned mixing. Following PiggyCast, we fit one XGBoost regressor [Chen and Guestrin, 2016] per lead. The forecast-only variant uses seven forecast anomalies and the six available ensemble-spread fields (DLESyM is deterministic). The full variant adds latitude, longitude, and sine/cosine of verification-midpoint day of year. A location/calendar-only ablation removes all forecast inputs. Early stopping on 2024 selects the tree count; a fresh model with that count is refit using 2020-2024 and applied once to 2025. Area weights are used in fitting and validation. *Lightweight* here means six compact tree models and no retraining of any forecasting system; we do not make an unmeasured compute or data-efficiency claim.

Scores and uncertainty. The primary score is case-level, area-weighted spatial anomaly correlation (ACC), averaged equally across initialization-week cases. Secondary scores are RMSE, MAE, signed bias, and wet-area fraction error at 1 mm day^{-1} . We form paired percentile intervals from 2,000 circular moving-block draws over ordered initialization dates, keeping all six leads from an initialization together. The primary block length is four initializations; lengths two and eight are sensitivity checks. These are descriptive effect intervals, not p -values.

4 Results

4.1 National average

Full PiggyCast has the highest mean ACC (0.347) and lowest RMSE (4.989) among the evaluated methods (Table 1). Relative to equal weighting, its paired ACC difference is 0.040 [0.011, 0.070], while RMSE changes by $-0.186 [-0.290, -0.096] \text{ mm day}^{-1}$. MAE falls from 3.715

to 3.603, and signed bias changes from -0.428 to $-0.303 \text{ mm day}^{-1}$. These jointly rule out an ACC-only improvement accompanied by worse rainfall-amplitude error.

Simple inverse-validation-RMSE weighting provides only a small point gain (0.313 versus 0.307 ACC). The forecast-only mixer reaches 0.339 ACC and $5.028 \text{ mm day}^{-1}$ RMSE. Its ACC difference from the location/calendar-only ablation is 0.238 [0.209, 0.266], establishing that forecast-state inputs carry useful information. This ablation does not, by itself, prove that every model contributes complementary information.

The stricter comparison remains uncertain. Full PiggyCast exceeds the per-lead 2024-RMSE-selected baseline by 0.052 mean ACC, but the interval $[-0.002, 0.093]$ overlaps zero. Accordingly, the predeclared headline gate for robust superiority to that baseline is closed, even though its point ACC, RMSE, and MAE are favorable.

4.2 Lead dependence

The national mean masks a sharp horizon boundary (Figure 2). Full PiggyCast changes ACC relative to equal weighting by 0.057, 0.064, 0.070, and 0.099 at Weeks 1-4, but by -0.001 and -0.047 at Weeks 5-6. Its RMSE is lower at every lead, including when its long-lead anomaly pattern is not better. Reporting only a six-week average would therefore conceal the main deployment limitation.

4.3 Regional heterogeneity

The full-minus-equal ACC difference is 0.128 [0.085, 0.177] in northwest India and 0.060 [0.018, 0.103] in the south peninsula. Central India is uncertain at 0.011 $[-0.036, 0.058]$, while east/northeast India shows a clear loss of $-0.097 [-0.175, -0.037]$ (Figure 3). Thus two regional intervals are positive, one is inconclusive, and one is negative. A national average cannot justify geographically uniform use.

5 Implications and limitations

The evidence supports a bounded forecast-quality conclusion: compact, forecast-conditioned mixing improves an uncalibrated equal-weight ensemble over India on average, with the clearest value at Weeks 1-4 and in northwest and south-peninsula regions. It does not establish uniformly better S2S prediction, operational value, or improved decisions. A cautious deployment would retain

Table 1: Mean 2025 India scores over 35 JJAS initializations and six leads. RMSE, MAE, and bias are in mm day⁻¹. The table retains only claim-relevant comparators; all systems and wet-area error are in the appendix.

Method	ACC $\uparrow$	RMSE $\downarrow$	MAE $\downarrow$	Bias
PiggyCast, full	0.347	4.989	3.603	-0.303
PiggyCast, forecast only	0.339	5.028	3.626	-0.288
Validation-weighted mean	0.313	5.150	3.700	-0.397
Equal-weight mean	0.307	5.174	3.715	-0.428
Per-lead 2024-RMSE-selected	0.296	5.283	3.845	0.157
ECMWF	0.305	5.240	3.822	0.253

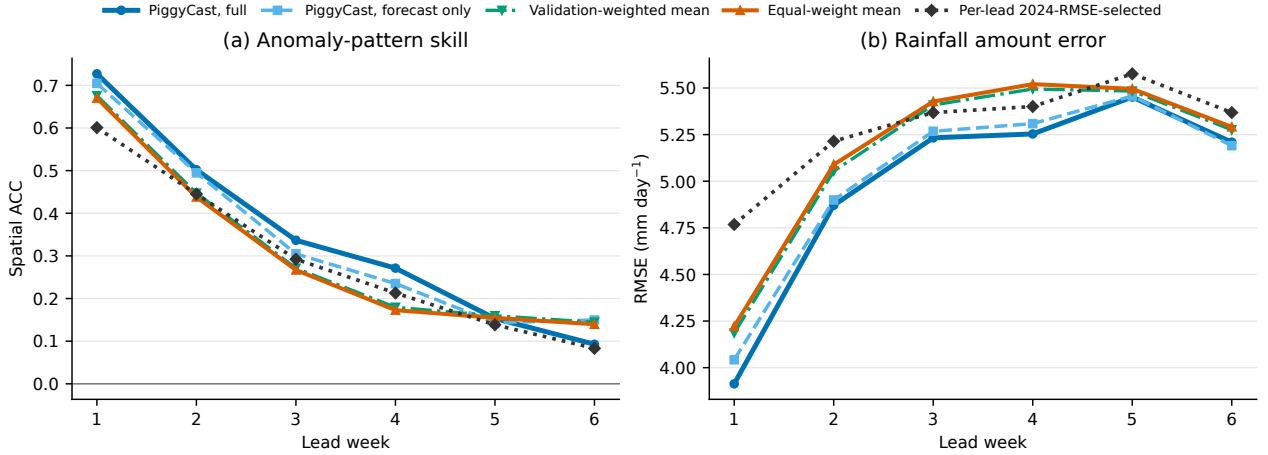


Figure 2: IMD-referenced India skill by lead. Curves are limited to the post-processing methods and pre-specified comparators needed to interpret the claim; the complete seven-system comparison is in Figure 4.

the original forecasts, monitor each lead and region, and abstain where validation does not support the mixer.

The evaluation contains only 35 initializations from one year, and serial dependence limits precision. The full-versus-selected lower interval endpoint changes from 0.009 at block length two to -0.002 and -0.002 at lengths four and eight; the equal-weight conclusion remains positive at all three lengths. The benchmark evaluates deterministic ensemble means, not probabilistic calibration, extremes, monsoon onset, or decision value. Ensemble sizes and model versions differ, and five common 2024 dates are missing.

IMD is the sole observational reference in the reported results, so gauge coverage, one 1991–2019 normal, and nonstationarity remain sources of uncertainty. Approximately 0.5% of full-PiggyCast’s area-weighted national support is negative under the pre-specified no-clipping policy; physical non-negativity should be handled in a future calibration study, not silently repaired after seeing test scores. Finally, a planned FuXi neural correction is excluded because its saved evaluator defines Week 1 as initialization+0–6 rather than the benchmark’s +1–7 endpoints. Comparing it would trade methodological variety for a one-day label error.

Climate relevance and responsible use. Rainfall guidance can affect risk-sensitive decisions and resource allocation. The regional failure is therefore not a cos-

metic subgroup result: it argues against presenting one national skill number as readiness for use across India. This study supplies forecast-quality evidence only. Any downstream application requires locally relevant thresholds, probabilistic reliability, cost-loss or impact evaluation, and communication of uncertainty.

6 Conclusion

On a common JJAS 2025 benchmark, lightweight PiggyCast mixing improves equal-weight multi-model rainfall forecasts in national ACC, RMSE, and MAE. The same experiment also supplies the boundary of that result: uncertainty against a per-lead selected baseline, no ACC gain at Weeks 5–6, and a clear east/northeast loss. The central lesson is therefore conditional rather than promotional: forecast-conditioned mixing can add value, but lead- and region-specific evaluation must determine where it is used.

Artifact availability. The anonymized supplement contains the protocol, coverage, case metrics, intervals, serialized models, generation code, and hashes. Its audit regenerates every aggregate score; raw reproduction additionally requires access to provider and IMD source data.

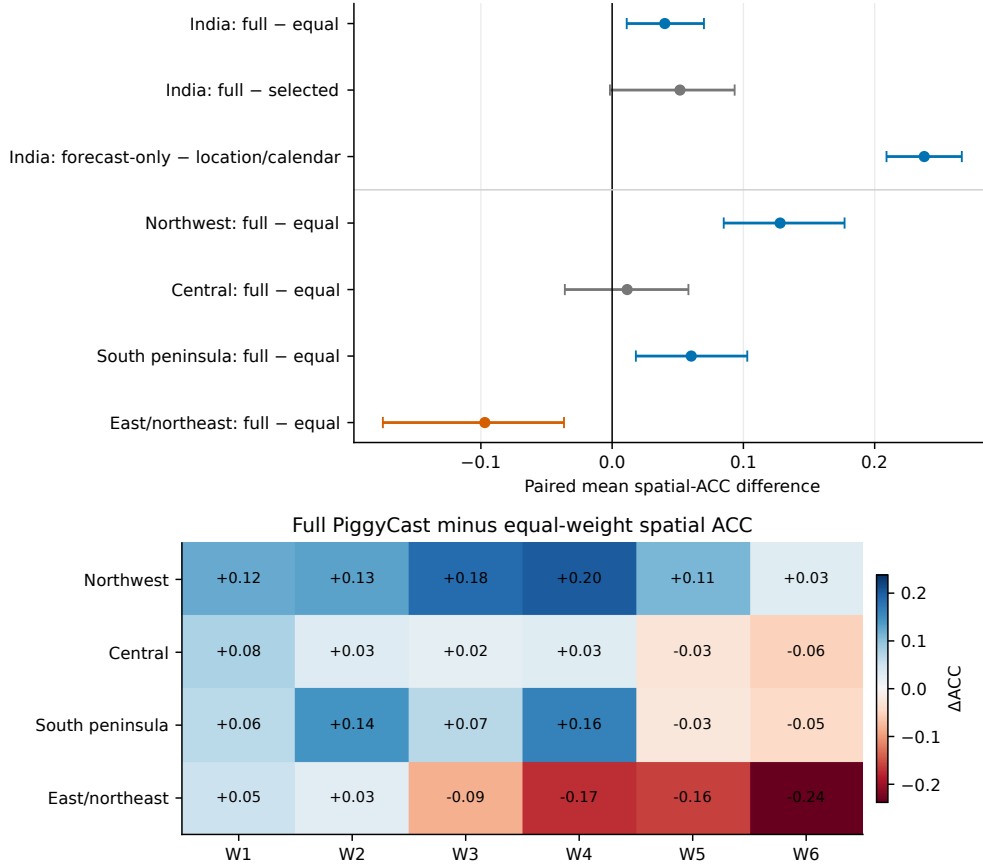


Figure 3: Failure-aware regional evaluation. Top: paired 2025 ACC differences with 95% circular moving-block percentile intervals. Blue intervals exclude zero positively, orange excludes zero negatively, and grey overlaps zero. The first three rows are pre-specified national contrasts; regional rows compare full PiggyCast with equal weighting. Bottom: the same full-minus-equal comparison by region and lead. Bottom-panel cells are descriptive point differences without per-cell intervals.

References

- Lei Chen, Xiaohui Zhong, Hao Li, Jie Wu, Bo Lu, Deliang Chen, Shang-Ping Xie, Qingchen Chao, Chensen Lin, Zixin Hu, and Yuan Qi. A machine learning model that outperforms conventional global subseasonal forecast models. *Nature Communications*, 15:6425, 2024. doi: 10.1038/s41467-024-50714-1.
- Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.
- Nathaniel Cresswell-Clay, Bowen Liu, Dale Durran, Andy Liu, Zachary I. Espinosa, Raul Moreno, and Matthias Karlbauer. A deep learning earth system model for stable and efficient simulation of the current climate. *arXiv preprint arXiv:2409.16247*, 2024. doi: 10.48550/arXiv.2409.16247.
- Josiah Kimani, Oliver Angélil, Chris Toumping, and Steffen Knoblauch. Piggycast—improving weather prediction accuracy through a stacking-based ensemble ai approach. In *NeurIPS 2025 Workshop on Tackling Climate Change with Machine Learning*, 2025. URL <https://www.climatechange.ai/papers/neurips2025/121>.
- Dmitrii Kochkov, Janni Yuval, Ian Langmore, Peter Norgaard, Jamie Smith, Griffin Mooers, Milan Klöwer, James Lottes, Stephan Rasp, Peter Düben, et al. Neural general circulation models for weather and climate. *Nature*, 632:1060–1066, 2024. doi: 10.1038/s41586-024-07744-y.
- D. S. Pai, Latha Sridhar, M. Rajeevan, O. P. Sreejith, N. S. Satbhai, and B. Mukhopadhyay. Development of a new high spatial resolution (0.25 degree x 0.25 degree) long period (1901–2010) daily gridded rainfall data set over india and its comparison with existing data sets over the region. *Mausam*, 65(1):1–18, 2014.
- Frédéric Vitart, Christelle Ardilouze, Ana Bonet, Anca Brookshaw, Mingyue Chen, Cristian Codorean, Michel Déqué, Laura Ferranti, Edmond Fucile, Manuel Fuentes, et al. The subseasonal to seasonal (s2s) prediction project database. *Bulletin of the American Meteorological Society*, 98(1):163–173, 2017. doi: 10.1175/BAMS-D-16-0017.1.

A Supplementary material

All reported appendix scores use IMD as the observational reference and the same dates, grid, anomaly normal, and fractional-area support as the main paper. No IMERG or temperature result is included in the current claims.

A.1 Forecast inventory and coverage

The five missing 2024 common initializations are June 6, 10, 13, 17, and August 5. Exact ISO dates are stored in `coverage.csv`; per-system availability, coordinates, units, and source metadata hashes are stored in `preflight/coverage_preflight.csv` and `preflight/preflight.json`. The metadata-only coverage preflight records `forecast_values_read=false`. DLESyM and NeuralGCM 2025 files enter through native-file adapters after the same coordinate, unit, and 42-period completeness checks as the standardized stores.

A.2 All systems and all lead weeks

Figure 4 replaces a visually crowded multi-line plot. It also keeps ACC and RMSE together: a method can reduce rainfall-amplitude error without improving the spatial anomaly pattern, as full PiggyCast does at the longest leads.

A.3 Regional and lead-wise behavior

The lead-by-region diagnostic is shown in Figure 3 of the main paper so that local lead-time failures are not hidden in the supplement.

Table 2: Forecast cohort. The 2024 common intersection contains 30 cases; 2020–2023 and 2025 contain 35 per year. Operational ensemble size changes where shown do not alter the ensemble-mean evaluation contract.

System	Type	Members	2020–23 cases/year	2025 cases
CMA operational	physics	4	35	35
DLESyM v0	AI Earth system	1	35	35
ECMWF operational	physics	51 or 101	35	35
FuXi-S2S	AI subseasonal	50	35	35
NCEP operational	physics	16	35	35
NeuralGCM	hybrid AI–physics	10	35	35
UKMO operational	physics	4	35	35

Table 3: Complete 2025 national score table. RMSE, MAE, and bias are in mm day^{-1} ; wet-area error is forecast minus observed weighted fraction at 1 mm day^{-1} . Negative support is the area-weighted fraction of evaluated support below zero under the no-clipping policy.

Method	ACC $\uparrow$	RMSE $\downarrow$	MAE $\downarrow$	Bias	Wet-area error	Negative support (%)
PiggyCast, full	0.347	4.989	3.603	-0.303	0.113	0.52
PiggyCast, forecast only	0.339	5.028	3.626	-0.288	0.117	0.16
Validation-weighted mean	0.313	5.150	3.700	-0.397	0.114	0.00
Equal-weight mean	0.307	5.174	3.715	-0.428	0.114	0.00
ECMWF	0.305	5.240	3.822	0.253	0.107	0.00
Per-lead 2024-RMSE-selected	0.296	5.283	3.845	0.157	0.097	0.00
FuXi-S2S	0.276	5.683	4.014	-0.363	0.047	0.00
NeuralGCM	0.236	5.694	4.094	-0.429	0.099	0.00
UKMO	0.226	6.370	4.522	0.328	0.044	0.00
CMA	0.197	6.852	4.839	-0.364	0.052	0.06
NCEP	0.154	6.421	4.538	-1.471	-0.013	0.02
DLESyM v0	0.102	6.735	4.815	-0.950	0.010	0.00
Location/calendar only	0.101	5.587	4.091	-0.402	0.144	0.06

Table 4: Full PiggyCast minus equal-weight regional ACC, averaged across initializations and leads.

Domain: full – equal weight	Mean	2.5th pct.	97.5th pct.
northwest india	0.128	0.085	0.177
central india	0.011	-0.036	0.058
south peninsula	0.060	0.018	0.103
east northeast india	-0.097	-0.175	-0.037

Table 5: 95% paired ACC intervals under alternative moving-block lengths. Point effects are 0.040 (full minus equal) and 0.052 (full minus per-lead selected) at every length.

Block length	Full – equal	Full – per-lead selected
2	[0.012, 0.067]	[0.009, 0.089]
4	[0.011, 0.070]	[-0.002, 0.093]
8	[0.017, 0.070]	[-0.002, 0.096]

A.4 Claim gates and uncertainty

The pre-specified headline requires positive lower 95% ACC endpoints against both equal weighting and the per-lead 2024-RMSE-selected baseline, a positive forecast-only versus location/calendar-only endpoint, at most 5% relative RMSE regression, at most 0.5 mm day^{-1} absolute-bias regression, and positive regional ACC point differences in at least three of four regions. All checks pass except the selected-baseline ACC endpoint. The regional point gate should not be read as evidence of uniform benefit: only northwest and south intervals exclude zero positively, while east/northeast excludes zero negatively. The machine-readable decision is in `gate_report.json`.

The equal-weight result is stable to the tested block lengths. The stronger selected-baseline comparison is sensitive at its lower endpoint, which motivates the conservative main-text interpretation.

A.5 Preprocessing and quality control

Daily precipitation is expressed as mm day^{-1} before weekly averaging. Each week requires seven finite daily periods; partial weeks are rejected. Latitude order, longitude convention, initialization timestamps, accumulation period endpoints, and ensemble dimensions are checked before scoring. The same IMD observation fraction and fractional polygon overlap enter every method’s area weights. ACC is undefined for a case with zero weighted spatial variance; such a case is recorded as missing rather than replaced by zero.

Small finite negative increments created when accumulated precipitation is differenced are retained. Full PiggyCast has 0.5% negative area-weighted national support overall. This choice avoids a test-informed clipping rule and exposes a calibration issue for future work.

A.6 Hyperparameters and compute

XGBoost uses squared-error loss, histogram trees on CPU, maximum depth 4, minimum child weight 5, learn-

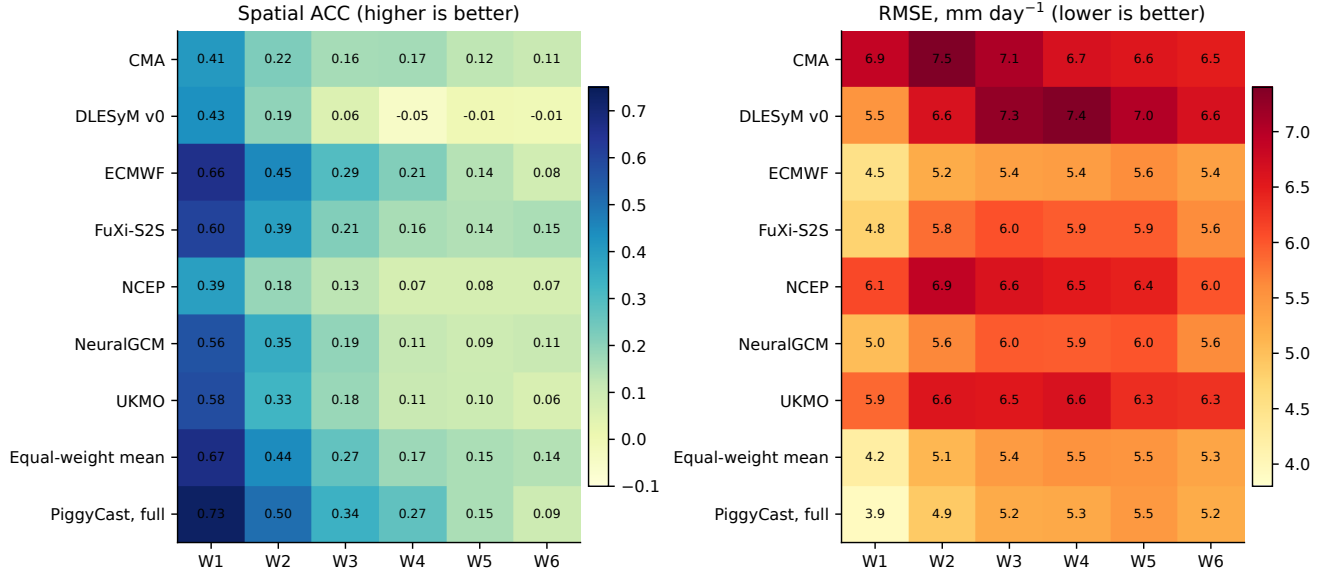


Figure 4: IMD-referenced 2025 national comparison for every base system, equal weighting, and full PiggyCast at Weeks 1–6. The defensible pattern is rapid, heterogeneous skill decay rather than a blanket statement that the systems are unskillful: several Week-1 forecasts have useful anomaly-pattern agreement, while Weeks 5–6 are consistently difficult.

ing rate 0.03, row subsampling 0.8, column subsampling 0.9, L_2 regularization 1, seed 42, and at most 600 trees with patience 50. Exact selected tree counts, feature names, weighted row counts, and serialized model paths for all 18 lead-variant combinations are stored in `fit_metadata.json`. Runtime and energy were not instrumented reliably, so no numerical efficiency or emissions claim is made.

A.7 Neural correction exclusion

The available FuXi neural-correction control defines Week 1 as initialization through initialization+6 days, while this benchmark defines daily lead 1 as the accumulation interval ending initialization+1 day. Using its saved model or scores would shift every weekly label by one day. It is therefore excluded from the common-date result and can be revisited only after training under the +1 through +42 endpoint contract. It would be a complementary alternative, not a correction applied after PiggyCast.

A.8 Reproduction commands

```
PYTHONPATH=src python run.py preflight
PYTHONPATH=src python run.py run
PYTHONPATH=src python run.py audit
MPLCONFIGDIR=/tmp/india-s2s-mpl \
  PYTHONPATH=src python make_paper.py
```

The runner refuses an existing output directory. A repeat must use a new path and is a distinct evaluation run, never a replacement selected by score. SHA256 values for saved scientific artifacts appear in `manifest.json`; manuscript-source and PDF hashes appear in `submission_manifest.json`.